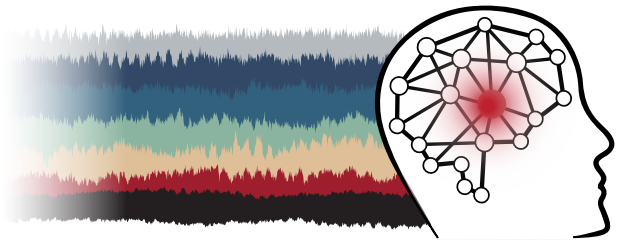
Dr. rer. nat.
Julian Q. Kosciessa

Thomas van Aquinostraat 4, Nijmegen

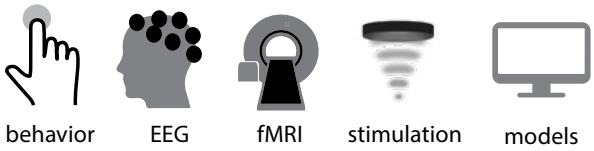
julian.kosciessa@donders.ru.nl

ORCID: 0000-0002-4553-2794



*I am a **cognitive computational neuroscientist** studying how brain dynamics keep perception and cognition adaptive in uncertain and changing environments. My research tests how thalamic circuits adjust the precision of cortical processing, combining experimental tasks and behavioural modelling with multimodal neuroimaging (EEG, fMRI) and image-guided transcranial ultrasound stimulation (TUS).*

I extend neruoscientific methods via open and FAIR scientific software development.



Research Experience

Postdoctoral Research Fellow — **2022 – PRESENT**
Donders Institute for Brain, Cognition and Behaviour
Nijmegen, Netherlands

Pre-/postdoctoral Research Fellow — **2016 – 2022**
IMPRS Comp2Psych
Max Planck UCL Centre for Computational Psychiatry and Aging Research
Berlin, Germany

2010 – 2016

Research Assistant/Intern
Berlin, Germany
London, UK
Singapore, Singapore

Education

Humboldt Universität zu Berlin — **2016 – 2020**
Psychology
Dr. rer. nat. (summa cum laude)

Humboldt Universität zu Berlin — **2014 – 2016**
Mind & Brain – Track Brain
M.Sc. Master of Science

Freie Universität Berlin — **2011 – 2014**
Psychology
B.Sc. Bachelor of Science

Comp. Skills

MATLAB
R
UNIX
Python
Git

Languages

German	native
English	near-native
Chinese	HSK5-6
Dutch	B1
French	A2

Key Publications

	Citations
Kosciessa, J. Q., Lindenberger, U., & Garrett, D. D. (2021) Thalamocortical excitability modulation guides human perception under uncertainty *Nature Communications*	**129**
Kosciessa, J. Q., Kloosterman, N. A., & Garrett, D. D. (2020) Standard multiscale entropy reflects neural dynamics at mismatched temporal scales: What's signal irregularity got to do with it? *PLoS Computational Biology*	**129**
Kosciessa, J. Q., Grandy, T. H., Garrett, D. D., & Werkle-Bergner, M. (2020) Single-trial characterization of neural rhythms: Potential and challenges. *NeuroImage*	**151**

RESEARCH

11/2023 - PRESENT **Research Fellow**
Donders Centre for Cognition
Radboud University, Nijmegen, The Netherlands
Cognitive Neurodynamics

09/2022 – PRESENT **Postdoctoral Researcher | Radboud Excellence Fellow**
Donders Institute for Brain, Cognition and Behaviour
Radboud University, Nijmegen, The Netherlands
Cognitive Neuromodulation | PI: Dr. Lennart Verhagen

07/2020 – 06/2022 **Postdoctoral Researcher**
Max Planck Institute for Human Development, Berlin, Germany

10/2016 – 03/2020 **Predoctoral Research Fellow**
IMPRS COMP2PSYCH: International Max Planck Research School on Computational Methods in Psychiatry and Ageing Research
Max Planck UCL Centre for Computational Psychiatry and Aging Research
Max Planck Institute for Human Development, Berlin, Germany
Supervisors: Prof. Dr. Ulman Lindenberger, Dr. Douglas D. Garrett

10/2015 – 03/2016 **Research Intern**
UCL Institute of Cognitive Neuroscience
PIs: Prof. Emrah Düzel & Prof. Ray Dolan
PI: Dr. Dorothea Hämmerer

03/2015 – 07/2015 **Research Intern**
Max Planck Institute for Human Development, Berlin, Germany
Center for Adaptive Rationality (ARC)
PI: Dr. Wouter van den Bos

09/2012 – 09/2013 **Research Assistant**
07/2014 – 09/2015 Max Planck Institute for Human Development, Berlin, Germany
04/2016 – 09/2016 Cognitive and neuronal dynamics of memory across the lifespan
PIs: Dr. Markus Werkle-Bergner & Dr. Yee Lee Shing

01/2014 – 05/2014 **Research Intern**
Cognitive Neuroscience Laboratory, Duke-NUS, Singapore
PI: Prof. Michael Chee
Supervisor: Dr. Irma Kurniawan

EDUCATION

10/2016 – 10/2020 **Humboldt Universität zu Berlin**
Psychology. Dr. rer. nat. (summa cum laude)

10/2014 – 09/2016 **Humboldt Universität zu Berlin**
Mind & Brain – Track Brain. M.Sc. Master of Science (GPA: 1.0)

09/2015 – 04/2016 **University College London**
Two Erasmus exchange terms. Institute of Neurology

07/2013 – 05/2014 **National University of Singapore (NUS)**
Two exchange semesters. Faculty of Arts and Social Sciences

10/2011 – 09/2014 **Freie Universität Berlin**
Psychology. B.Sc. Bachelor of Science (GPA: 1.1)

Teaching

2023–2026 | Lab Rotations, MSc Cognitive Neuroscience, Radboud University.
2023–2025 | Practical Training and Thesis. MSc Cognitive Neuroscience. Radboud University.
2025 | Lecture: Flexible Perceptual and Cognitive Control. Trends in Cognitive Neuroscience, MSc Cognitive Neuroscience, Radboud University.
2025 | Lecture: Managing Uncertainty in Perceptual and Cognitive Control. Ghent University.
2024 | Lecture: Evolving Models of Flexible Neuro-Cognitive Control. Trends in Cognitive Neuroscience, MSc Cognitive Neuroscience, Radboud University.
2024-2026 | Workshop: Multimodal TUS-fMRI. Donders Toolkit Advanced (f)MRI. RU.
2023 | Workshop: Managing Your Data with DataLad. CuttingEEG, Frankfurt.
2023–2026 | Workshop: Simulations for Transcranial Ultrasound Stimulation. NeuroTechEU Course / Donders NIBS Toolkit. Radboud University. [link]
2023 | Seminar Teacher: Higher-Order Cognition and Emotion. Radboud UMC.
2023 | Psychology Research Project 3. Teacher. Radboud University
2022 | Workshops: Reusable data management with DataLad.
Cognitive Psychology. University of Münster.
Donders Centre for Cognitive Neuroimaging. Radboud University.
2020 | Workshop: Multi-scale entropy as a tool to characterize neural signal irregularity. EEG Meeting. Max Planck Institute for Human Development. Berlin.
2018 | Seminar: Methods for the analysis of rhythmic and arrhythmic brain activity.International Max Planck Research School on the Life Course. Berlin.

Talks (selected)

2027 | Invited Symposium Talk: Thalamic contributions to decisions under uncertainty. ThalNet Conference. San Sebastián, Spain (19-21 May 2027)
2026 | Invited Keynote Lecture: Impact of acoustic property uncertainty on transcranial ultrasound simulation. Focused Ultrasound Neuromodulation Conference (FUN26). Paris, France
2026 | Invited Symposium Talk: Neuro-computational signatures of flexible decisions under uncertainty. ESCoP-CoPM 2026. Hangzhou, China
2025 | Invited Symposium Talk: Neuromodulatory Influences On Aging Cognition: Novel Measurement And Intervention Approaches In Health And Disease. ICON2025. Porto, Portugal
2025 | Talk: Biasing perceptual and cognitive control with thalamic ultrasound stimulation. Focused Ultrasound Neuromodulation Conference (FUN25), Hong Kong
2024 | Talk: Neuroscientific data management using DataLad. distribits conference. Düsseldorf. Germany [link]
2023 | Invited Keynote Talk: Signs, signals, and noise in human brain dynamics. CuttingEEG conference. Frankfurt. Germany [link]
2022 | Invited Research Talks: Dynamic neural regimes for flexible decisions under uncertainty. Translational Decision-Making Seminar. University of Minnesota/Université de Montréal. Biopsychology. University of Münster. Germany
2022 | Invited Symposium Talk: Influences of arousal and cortical excitability on adaptive perceptual decision making. ICON 2022. Helsinki, Finland
2021 | Research Talk: The role of neural dynamics in flexible perception under uncertainty. Computational Neuroscience Symposium. Osnabrück, Germany
2021 | Invited Research Talks: Thalamocortical excitability adjustments guide human perception under uncertainty. Shine Lab, University of Sydney, Australia / Halassa Lab, Massachusetts Institute of Technology (MIT), U.S.A.
2020 | Invited Colloquium Talk: Measurement and relevance of rhythmic and aperiodic human brain dynamics. Technical University. Berlin, Germany

Fellowships

2022 – 2024 Radboud Excellence Initiative Fellowship
Donders Centre for Cognition, Radboud University, The Netherlands
2022 ReproNim/INCF Fellowship (https://repronim.org/fellowship/)
2016 – 2020 IMPRS COMP2PSYCH Fellowship
International Max Planck Research School on Computational Methods in Psychiatry and Ageing Research, Germany

Awards

2026 | Donders Open Research Output Award
2025 | Poster Award (Gold) | Focused Ultrasound Neuromodulation Conference (FUN25)
2022 | Radboud Excellence Initiative Fellowship (280.000 EUR incl. in-kind contributions)
2022 | Otto Hahn Medal of the Max Planck Society (7.500 EUR)
2022 | Conference Travel Grant | DAAD | International Conference of Cognitive Neuroscience
2021 | DGPA Brain Products Young Scientist Award 2021
2021 | Conference Travel Grant | DAAD | OHBM Meeting 2021
2021 | Merit Abstract Award | OHBM Meeting 2021
2018 | IBRO Poster Award | Interpreting BOLD 2018
2018 | Conference Travel Grant | DAAD | Interpreting BOLD 2018 (Oxford, UK)
2015 | Erasmus Stipend (2015/16) | DAAD | University College London, UK
2014 | PROMOS Stipend | DAAD | National University Singapore, Singapore

Mentorship

2025 | Ruggero Condorelli | Research Assistant (MSc)
2026+ PhD student, Portugal
2025/26 | Narges Naghibi | MSc Cognitive Neuropsychology, Vrije Universiteit Amsterdam
2025/26 | Lena Grondelaers | MSc Cognitive Neuroscience, Radboud University
2025/26 | Javier Gómez Pérez | MSc Cognitive Neuroscience, Radboud University
2024/25 | Joana Pedroso de Faria | MSc Cognitive Neuroscience, Radboud University
2024/25 | Sebastian Reichstein | MSc Cognitive Neuroscience, Radboud University
2026+ PhD student at FernUni Schweiz, Brig, Switzerland
2023/24 | Martin Wimmers | MSc Cognitive Neuroscience, Radboud University
2025+ PhD student at TU Eindhoven
2023 | Jesse Lam | MSc Cognitive Neuroscience, Radboud University
2023 | Lieke Hendrix | BSc Biology, Radboud University
2023 | Sara Mulders | BSc Psychology, Radboud University
2023 | Siem van der Sluijs | BSc Psychology, Radboud University
2021/22 | Claire Pleche | M.Sc. Cognitive Neuroscience, Ecole Normale Supérieure de Paris, France
2022+ PhD student at Donders Institute
2021-2023 | Mentor at Neuromatch Academy

Ultrasound Trials *(CCMO)*

NL86115.091.24 | Biasing decision flexibility with thalamic ultrasound stimulation [link]
NL83639.091.23 | Imaging deep target engagement of transcranial ultrasonic stimulation [link]

Professional Activities & Leadership

Ad-hoc peer review: *PNAS, TICS, PLoS Biology, NeuroImage (13x), NeuroImage: Clinical, Journal of Neuroscience, Cerebral Cortex, Journal of Neurophysiology, Communications Biology, Psychophysiology, International Journal of Psychophysiology, Brain Topography, European Journal of Neuroscience, Nonlinear Dynamics, Mindfulness, PLoS One*

2027 | Cognitive Computational Neuroscience (CCN) | Communications Committee (Advising Chair)
2026 | Cognitive Computational Neuroscience (CCN) | Communications Committee (Chair)
2026 | AAPG [French National Research Agency] (CE37: Integrative and Cognitive Neuroscience) | Reviewer
2025 | Cognitive Computational Neuroscience (CCN) | Communications Committee (Rising Chair)
2025 | Donders Strategy Evaluation (2020-2024) | Postdoc representative
2025 | AAPG [French National Research Agency] (CE37: Integrative and Cognitive Neuroscience) | Reviewer
2025+ | ITRUSST Acoustic Properties Working Group | Member
2024 | Hiring Committee PostDoc Coordinator | Member
2022+ | Co-founder & -organizer Donders PostDoc Ponders
2022-2024 | Donders Institute PostDoc Council | Member
2022-2024 | PostDoc representative Donders Centre for Cognition
2022/23 | Co-founder & co-organizer: "DCC Nexus" meeting series
2022+ | International Transcranial Ultrasonic Stimulation Safety and Standards (ITRUSST) | Member
2021+ | International Neuroinformatics Coordinating Facility (INCF) | Member
2020+ | Deutsche Gesellschaft für Psychology (DGPs) | Associate Member
2018+ | Organization for Human Brain Mapping (OHBM) | Member

Publications (*corresponding author)

10 Skowron, A., **Kosciessa, J. Q.,** Lorenz, R. C., Hertwig, R., van den Bos, W., & Garrett, D. D. (2025). Neural variability in the default mode network compresses with increasing belief precision during Bayesian inference. Cerebral Cortex, 35(9).

46 Mooraj, Z., Salami, A., Campbell, K. L., Dahl, M. J., **Kosciessa, J. Q.**, Nassar, M. R., Werkle-Bergner, M., Craik, F. I. M., Lindenberger, U., Mayr, U., Rajah, M. N., Raz, N., Nyberg, L., & Garrett, D. D. (2025). Toward a functional future for the cognitive neuroscience of human aging. Neuron, 113(1), 154-183.

12 **Kosciessa, J. Q.*,** Mayr, U., Lindenberger, U., & Garrett, D. D. (2024). Broadscale dampening of uncertainty adjustment in the aging brain. Nature Communications, 15(1), 10717.

129 **Kosciessa, J. Q.***, Lindenberger, U., & Garrett, D. D. (2021). Thalamocortical excitability modulation guides human perception under uncertainty. *Nature Communications, 12(1), 2430*.

36 Kloosterman, N. A., **Kosciessa, J. Q.**, Lindenberger, U., Fahrenfort, J. J., & Garrett, D.D. (2020). Boosts in brain signal variability track liberal shifts in decision bias. *Elife, 9*.

129 **Kosciessa, J. Q.***, Kloosterman, N. A., & Garrett, D. D. (2020). Standard multiscale entropy reflects neural dynamics at mismatched temporal scales: What's signal irregularity got to do with it? *PLoS Computational Biology, 16(5)*.

151 **Kosciessa, J. Q.***, Grandy, T. H., Garrett, D. D., & Werkle-Bergner, M. (2020). Single-trial characterization of neural rhythms: Potential and challenges. *NeuroImage, 206, 116331*.

205 Hämmerer, D., Callaghan, M. F., Hopkins, A., **Kosciessa, J.**, Betts, M., Cardenas-Blanco, A., Kanowski, M., Weiskopf, N., Dayan, P., Dolan, R. J., & Düzel, E. (2018). Locus coeruleus integrity in old age is selectively related to memories linked with salient negative events. Proceedings *of the National Academy of Sciences of the United States of America, 115, 2228-2233*.

Software / Tools (selected)

PRESTUS | PREprocessing & Simulations for Transcranial Ultrasound Stimulation | 10.5281/zenodo.15095860
eBOSC | extended Better OSCillation Detection | 10.5281/zenodo.4668501
eBOSC (python) | extended Better OSCillation Detection (python version) | 10.5281/zenodo.5898812
mMSE | modified multiscale entropy | 10.5281/zenodo.4672137
MAAT | Multi-attribute attention task | 10.5281/zenodo.14216065
CARL | Computer-Assisted Response Labeler | 10.5281/zenodo.7505621

PUBLICATION LIST

Journal Publications *(*corresponding author)*

- Skowron, A., **Kosciessa, J. Q.**, Lorenz, R. C., Hertwig, R., van den Bos, W., & Garrett, D. D. (2025). Neural variability in the default mode network compresses with increasing belief precision during Bayesian inference. Cerebral Cortex, 35(9).

  BOLD variability tracks changes in internal uncertainty levels during a latent inference task.

  10 article

- Mooraj, Z., Salami, A., Campbell, K. L., Dahl, M. J., **Kosciessa, J. Q.**, Nassar, M. R., Werkle-Bergner, M., Craik, F. I. M., Lindenberger, U., Mayr, U., Rajah, M. N., Raz, N., Nyberg, L., & Garrett, D. D. (2025). Toward a functional future for the cognitive neuroscience of human aging. Neuron, 113(1), 154-183.

  A perspective on dominant methodologies in the cognitive neuroscience of human aging, identifying an overreliance of structural and resting-state measures. The perspective advocates for a more task-centric, computationally-informed, approach in the field.

  46 article

- **Kosciessa, J. Q.*,** Mayr, U., Lindenberger, U., & Garrett, D. D. (2024). Broadscale dampening of uncertainty adjustment in the aging brain. Nature Communications, 15(1), 10717.

  Task uncertainty is tracked by an integrated neural response (EEG & fMRI-based) across 100 younger and older adults, including prefrontal-thalamic modulation. This response dampens in older age, while response maintenance is associated with higher cognitive flexibility.

  12 article data code

- **Kosciessa, J. Q.***, Lindenberger, U., & Garrett, D. D. (2021). Thalamocortical excitability modulation guides human perception under uncertainty. *Nature Communications, 12(1), 2430*.

  Higher-order thalamic activation increases when contextual uncertainty ambiguates which environmental features are critical for an upcoming choice, and is associated with switches from a rhythmic to an aperiodic processing mode.

  129 article data code task press release

- Kloosterman, N. A., **Kosciessa, J. Q.**, Lindenberger, U., Fahrenfort, J. J., & Garrett, D.D. (2020). Boosts in brain signal variability track liberal shifts in decision bias. *Elife, 9*.

  The magnitude of adaptive shifts from conservative to liberal decision biases under speed-accuracy emphasis is tracked by increasing signal variability in frontal cortex.

  36 article data code

- **Kosciessa, J. Q.***, Kloosterman, N. A., & Garrett, D. D. (2020). Standard multiscale entropy reflects neural dynamics at mismatched temporal scales: What's signal irregularity got to do with it? *PLoS Computational Biology, 16(5)*.

  Highlights and exemplifies biases in prior research using an information theoretic metric of signal irregularity and proposes avenues to adjudicate such issues in future applications.

129

article

data

code

toolbox

- **Kosciessa, J. Q.***, Grandy, T. H., Garrett, D. D., & Werkle-Bergner, M. (2020). Single-trial characterization of neural rhythms: Potential and challenges. *NeuroImage, 206, 116331*.

  Introduces a novel algorithm that separates neural rhythms from background activity in magnitude, space and time, and enables specific rhythm characterization when boundary conditions are met.

151

article

code

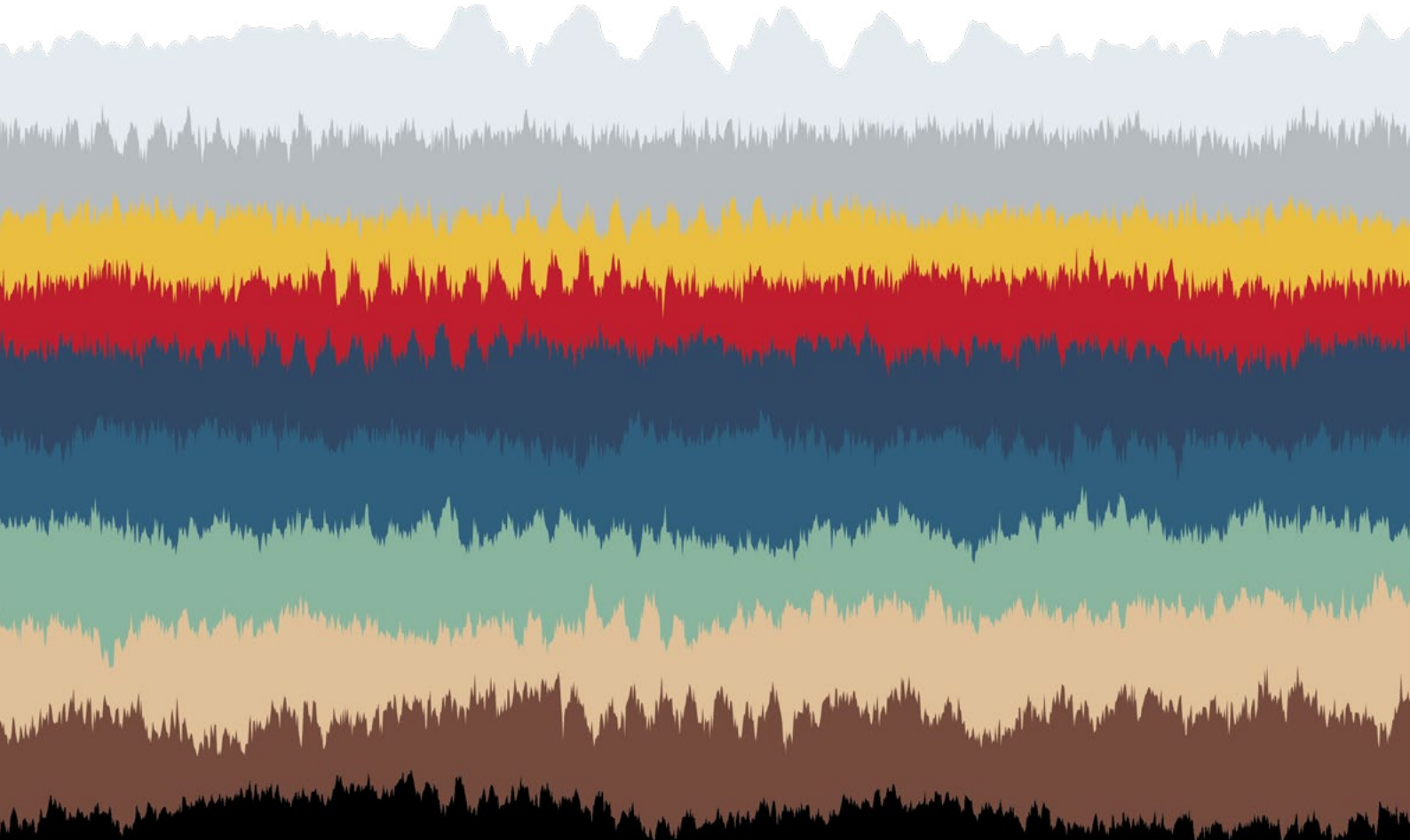
toolbox

- Hämmerer, D., Callaghan, M. F., Hopkins, A., **Kosciessa, J.**, Betts, M., Cardenas-Blanco, A., Kanowski, M., Weiskopf, N., Dayan, P., Dolan, R. J., & Düzel, E. (2018). Locus coeruleus integrity in old age is selectively related to memories linked with salient negative events. *Proceedings of the National Academy of Sciences of the United States of America, 115, 2228-2233*.

  Quantitative imaging indicates structural reductions in brainstem locus coeruleus integrity with increasing adult age, and links related noradrenergic drive to the encoding of salient events.

205

article



Monographs/Theses

- **Kosciessa, J. Q.** (2020, Dr. rer. nat.). Measurement and relevance of rhythmic and aperiodic human brain dynamics. *Humboldt-Universität zu Berlin*.

  This dissertation highlights improvements in the ability to selectively characterize rhythmic and aperiodic fluctuations, and discusses potential generating mechanisms as well as modulatory influences to contextualize their interpretation at the latent level of human brain function.

  article

- **Kosciessa, J. Q.** (2016, M. Sc.). Effects of short-term memory load and task training on the amplitude and abundance of rhythmic neural activity. *Humboldt-Universität zu Berlin*

  Neural rhythms are dominantly characterized by their power, but this measure conflates amplitude and duration. This work separates these two parameters of human alpha rhythms and investigates how they are modulated during working memory.

- **Kosciessa, J.** (2014, B. Sc.). The assessment of microsaccades from the rEOG. *Freie Universität Berlin*

  Microsaccades are small, high-velocity eye movements. This work explores the potential to use visual EEG channels to detect microsaccades without an eye tracker, and describes adult age differences in microsaccade characteristics.

Preprints / Conference Papers

- **Kosciessa, J. Q.***, Webb, T. D., Martin, E., Treeby, B., Butts Pauly, K., Verhagen, L., & Pichardo, S. (2026). Impact of acoustic property uncertainty on transcranial ultrasound simulation. bioRxiv. https://doi.org/10.64898/2026.10.30.763065

  preprint | code + data

- **Kosciessa, J. Q.***, Wimmers, M., Evans, C., Pedroso de Faria, J., Reichstein, S., Gómez Pérez, J., Grondelaers, L., Condorelli, R., & Verhagen, L. (2026). Dissecting thalamic uncertainty management with transcranial ultrasound stimulation. 9th Conference on Cognitive Computational Neuroscience (CCN 2026), New York City, USA. https://openreview.net/pdf?id=8v8K3ORKrw

  preprint

- Cesnaite, E., Algermissen, J., Vinding, M. C., Vitale, A., Pascarella, A., Fischer, N. L., Yang, Y.-F., Trübutschek, D., Gianelli, C., Marshall, C. R., … **Kosciessa, J. Q.**, … Nilsonne, G., & Busch, N. A. (2026). Many pipelines, one dataset: Analytic flexibility and its impact in EEG research. MetaArXiv. https://doi.org/10.31222/osf.io/wvx48_v2

  preprint | contribution | code + data

- Cesnaite, E., Vitale, A., Pascarella, A., Vinding, M. C., Algermissen, J., Fischer, N. L., Puoliväli, T., Gianelli, C., Marshall, C. R., Trübutschek, D., … **Kosciessa, J. Q.,** … Nilsonne, G., & Busch, N. A. (2025). The EEGManyPipelines dataset: Metascientific data on 168 independent analyses of a single EEG dataset. MetaArXiv. https://osf.io/preprints/metaarxiv/c4xwg_v1

  preprint

- **Kosciessa, J. Q.***, Wimmers, M., Krijgsman, S., & Verhagen, L. (2025). Dissociable effects of uncertainty in perceptual and cognitive control. 8th Conference on Cognitive Computational Neuroscience (CCN 2025), Amsterdam, The Netherlands.

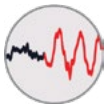
  preprint

- Garrett, D. D., Kloosterman, N. A., Epp, S., Chopurian, V., **Kosciessa, J. Q.**, Waschke, L., Skowron, A., Shine, J. M., Perry, A., Salami, A., Rieckmann, A., Papenberg, G., Wåhlin, A., Karalija, N., Andersson, M., Riklund, K., Lövdén, M., Bäckman, L., Nyberg, L., & Lindenberger, U. (2022). Dynamic regulation of neural variability during working memory reflects dopamine, functional integration, and decision-making. bioRxiv, 2022.2005.2005.490687. https://doi.org/10.1101/2022.05.05.490687

  preprint

Software / Tools / Data

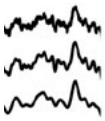
(abridged; see https://jkosciessa.github.io/tools for complete and updated list)



extended Better OSCillation Detection (**eBOSC**)

Toolbox to detect the occurrence of rhythms in (neural) time series. *MATLAB/Python*

DOI 10.5281/zenodo.4668502
DOI 10.5281/zenodo.5898813



modified multiscale entropy (**mMSE**)

Information-theoretic algorithm for the characterization of (neural) time series irregularity.

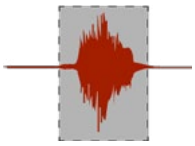
DOI 10.5281/zenodo.4672138



Multi-attribute attention task (**MAAT**)

Adaptation of the random dot motion paradigm, a key task in systems neuroscience.

DOI 10.5281/zenodo.14216065



Computer-Assisted Response Labeler (**CARL**)

Graphical User Interface that accelerates verbal response labelling in psychological experiments.

DOI 10.5281/zenodo.7505622



PREprocessing & Simulations for Transcranial Ultrasound Stimulation (**PRESTUS**)

Personalized ultrasound planning platform based on structural MRI images.

Gold Poster Award, FUN25; Donders Open Research Output Award 2026

DOI 10.5281/zenodo.15095860

Examples of integrated code & data DataLad datasets:
https://gin.g-node.org/juliankosciessa/eegmp_analysis
https://git.mpib-berlin.mpg.de/LNDG/stateswitchage/stsw
https://gin.g-node.org/StateSwitch/stsw



see also:
https://git.mpib-berlin.mpg.de/LNDG/stateswitch
https://osf.io/ug4b8/